Distilling VLM-Derived Indoor Affordance Scores from Synthetic Scene Images

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Abstract

Affordances—actions made possible by the features of a space—are fundamental to design. Thus, the ability to evaluate the design of a built environment depends on understanding the actions it affords. Affordance judgments depend on object identity, layout, clearance, and scene context, but obtaining dense human ratings is expensive. We therefore study a practical proxy question: how accurately can lightweight student models reproduce vision-language-model-derived affordance scores on synthetic indoor scenes? For our pilot study, using 420 Hypersim renderings held out by scene, we queried Qwen2-VL-7B for numeric 7-point scores and structured indicator checklists for five activities selected from an expert-curated taxonomy: Sleep, Cook, Computer Work, Casual Conversation, and Yoga/Stretching. We then trained student models on either raw RGB images or structured Mask2Former-derived scene features. On the held-out test scenes, the best CNN checkpoint per affordance reached macro RMSE 1.120, a mean-prediction baseline reached 1.154, a linear regression baseline reached 1.592, a LightGBM student on 310 structured features reached 0.977, and an indicator-augmented LightGBM student reached 0.765. Relative to Model B, Model D reduced RMSE on every affordance, with statistically significant gains on four of five tasks and the largest improvement on L091 Computer Work (ΔRMSE $= -0.328$, 95% bootstrap CI $[-0.500, -0.154]$). A permutation control shows that indicator semantics drive most of the gain on L059, L079, and L091, though dimensionality contributes on L130 and L141. Under an auxiliary binary view defined by score ≥ 4, Model D also attains the best macro F1 among the reproducible per-image baselines (0.679). We frame these results as evidence about distillation fidelity: lightweight tabular students can approximate the VLM teacher substantially better than image-only baselines on this synthetic benchmark, but correlation with human affordance ratings remains untested.

1 Introduction

Gibson's theory of affordances describes environments in terms of the actions they make possible [5]. For indoor scenes, that means a room may afford sleeping, cooking, working, socializing, or stretching depending on a combination of furniture and layout. Predicting such judgments from images is relevant to embodied AI, robotics, and functional scene understanding.

Recent vision-language models (VLMs) can make holistic scene judgments directly [1, 13], and recent annotation work has shown that large models can serve as useful proxy labelers in some domains when human annotation is scarce [6]. Our paper adopts that paradigm, but with an explicit caveat: Qwen2-VL-7B is a scalable *proxy annotator*, not ground truth. Accordingly, our evaluation measures how well lightweight students reproduce the teacher's outputs, not whether those outputs match human affordance judgments. Making that distinction explicit is central to the paper.

Within that scope, we study whether structured VLM supervision is more useful than scalar targets alone. Qwen2-VL-7B produces both a 1–7 affordance score and an indicator checklist; the students then attempt to predict the scalar score from either RGB images or structured scene features. This yields three questions. First, how well can small students approximate VLM-derived scores on held-out scenes? Second, do structured features outperform image-only CNN baselines in this low-data setting? Third, does indicator-level distillation help beyond the scalar target alone?

Our main findings are narrower than the original draft claimed, but stronger in scientific terms. On a scene-level split of 420 Hypersim images, structured LightGBM students outperform the best retained CNN checkpoints and the mean baseline. Adding indicator features improves every affordance and is statistically significant on four of five tasks, though the L141 Yoga/Stretching gain is uncertain at this sample size. A random-indicator control shows that the improvement is not explained purely by adding more binary columns. We therefore present this work as an end-to-end distillation pipeline for VLM-derived indoor affordance scoring on synthetic scenes, not as a claim that absolute affordance prediction has been solved. For design and interaction research, a coarse classifier of what an interior appears to afford is useful even before full human validation: it supports archive-scale queries, comparison of design variants, and fast screening of scenes for deeper review. That is also the intended role in our broader Image_Tagger_3.5 system, which already extracts design-relevant visual, spatial, and cognitive attributes from interior images;

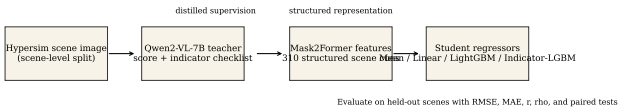


Figure 1: Two-phase workflow. Hypersim scene images are labeled by the VLM teacher, represented with structured scene features, and used to train lightweight students evaluated on held-out scenes.

affordance models would become one more analysis module in that practical toolchain.

2 Related Work

This project sits at the intersection of affordance modeling, distillation, and model-assisted annotation. Prior affordance work ranged from object-centric prediction [3, 4] to environment-scale reasoning in egocentric and 3D indoor settings [11, 12]. Our setting uses continuous room-level suitability scores from a VLM teacher.

The learning setup is response-based distillation [9] with an additional structured-rationale channel related to broader distillation surveys [7]. Hypersim [14], Mask2Former [2], LightGBM [10], and ResNet [8] are established components; the contribution is their combination into a reproducible scene-level VLM distillation pipeline. Our annotation strategy also follows work showing that large models can sometimes serve as useful proxy annotators, while still requiring careful caveats about construct validity [6].

3 Method

3.1 Data, Teacher Labels, and Split Strategy

The dataset contains 420 *synthetic* Hypersim images sampled across seven room clusters (bathroom, bedroom, dining room, kitchen, living room, lobby/lounge, and office), yielding 2,100 image-affordance rows after scoring each image on five affordances. The target affordances are L059 Sleep, L079 Cook, L091 Computer Work, L130 Casual Conversation, and L141 Yoga/Stretching. Qwen2-VL-7B assigns each image-affordance pair a scalar score on a 1–7 ordinal scale and emits a structured list of supporting or contradicting indicators.

Figure 1 summarizes the full workflow. We extracted 310 structured scene features from Mask2Former segmentations: 133 object-presence features, 133 counts, 15 pairwise 2D distances, 15 pairwise depth differences, and 14 aggregate scene statistics such as free-floor fraction, clutter, and depth spread. Indicator vocabulary consolidation yielded 624 canonical indicator types, expanded into 1,248 affordance-specific positive/negative binary columns.

The split is *scene-level* and stratified by room cluster: 224 train scenes, 32 validation scenes, and 64 test scenes, corresponding to 298/38/84 images per affordance. Grouping all camera views from the same scene into one split prevents leakage through shared geometry and layout.

Table 1: Dataset summary for the 420-image synthetic Hypersim benchmark. Scores are VLM-derived labels on a 1–7 scale.

Affordance	Mean	Std	Min	Max
L059 Sleep	3.150	1.334	1	7
L079 Cook	1.738	1.325	1	6
L091 Computer Work	3.848	0.999	1	6
L130 Conversation	4.481	1.126	1	7
L141 Yoga/Stretching	3.269	1.181	1	7
Images	420 synthetic renderings; 60 per room cluster			
Splits	224/32/64 scenes = 298/38/84 images per affordance			
Features	310 structured scene features + 1,248 indicator binaries			

3.2 Students and Training Details

We evaluate five predictors. The **Mean** baseline predicts the training-set mean score for every test image. The **Linear** baseline fits ordinary least squares on the 310 structured features. The **Best CNN** row reports the best retained ResNet-18 or ResNet-50 checkpoint per affordance from the original training sweep. **Model B** is a LightGBM regressor trained on the 310 structured features, and **Model D** augments Model B with the affordance-specific indicator columns.

For LightGBM, we reuse the best per-affordance Optuna configurations from the original experiments and retrain on the combined train+validation split with the standard L2 regression objective. The original search used 100 Optuna trials with 5-fold cross-validation over standard tree-depth, leaf-count, learning-rate, estimator-count, sampling, and regularization ranges.

4 Experiments

4.1 Primary Comparison and Baselines

Table 2 reports the main accuracy comparison. The key result survives the new baselines and the stricter framing: Model D is the strongest student on every affordance, with macro RMSE 0.765 and macro Pearson r 0.724. Model B is also clearly useful overall (macro RMSE 0.977), but its failure on L091 is now easier to interpret because the mean baseline exposes it directly: Model B's L091 RMSE (1.140) is worse than predicting the training-set mean (0.980), so on that affordance the raw structured features alone do not recover the teacher's signal.

This is not a pure architecture comparison between CNNs and gradient boosting. The CNNs receive raw pixels, whereas the tabular models receive a strong segmentation-derived representation. The supported claim is therefore that *structured feature pipelines outperform image-only fine-tuning in this low-data synthetic regime*, not that boosted trees are inherently superior to CNNs. The absolute error also remains large

Table 2: Test-set error metrics. RMSE and MAE are measured in score points on the 1–7 VLM scale. 'Best CNN' reports the best retained ResNet checkpoint per affordance.

	Mean		Linear		Best CNN		Model B		Model D	
Affordance	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE
L059 Sleep	1.099	0.876	1.088	0.798	1.096	0.906	0.814	0.658	**0.676**	**0.551**
L079 Cook	1.287	1.085	1.535	0.908	1.244	1.035	0.778	0.545	**0.523**	**0.279**
L091 Computer Work	**0.980**	0.637	2.169	1.268	0.958	0.669	1.140	0.791	0.812	**0.585**
L130 Conversation	1.180	1.010	1.189	0.946	1.137	1.000	0.978	0.790	**0.813**	**0.626**
L141 Yoga/Stretching	1.226	0.950	1.979	1.192	1.166	0.898	1.175	0.912	**1.002**	**0.793**
Macro average	1.154	0.912	1.592	1.023	1.120	0.902	0.977	0.739	**0.765**	**0.567**

Table 3: Test-set correlation metrics. The mean baseline has zero variance and therefore zero correlation by construction.

	Mean		Linear		Best CNN		Model B		Model D	
Affordance	r	ρ	r	ρ	r	ρ	r	ρ	r	ρ
L059 Sleep	0.000	0.000	0.656	0.731	0.213	0.284	0.696	0.653	**0.794**	**0.765**
L079 Cook	0.000	0.000	0.428	0.439	0.258	0.191	0.802	0.593	**0.924**	**0.729**
L091 Computer Work	0.000	0.000	-0.107	-0.068	0.254	0.255	-0.156	-0.063	**0.595**	**0.563**
L130 Conversation	0.000	0.000	0.514	0.524	0.302	0.291	0.587	0.580	**0.724**	**0.731**
L141 Yoga/Stretching	0.000	0.000	0.163	0.281	0.306	0.132	0.403	0.403	**0.582**	**0.527**
Macro average	0.000	0.000	0.331	0.381	0.266	0.231	0.466	0.433	**0.724**	**0.663**

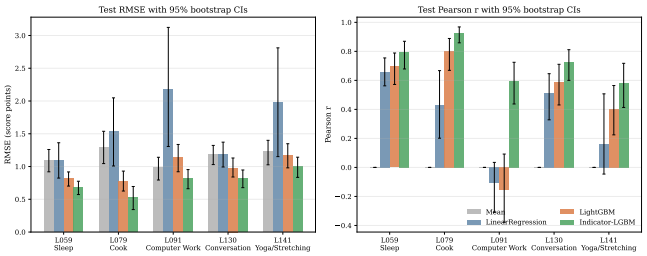


Figure 2: Primary reproducible baselines. Error bars show 95% bootstrap confidence intervals for RMSE and Pearson r for the mean, linear, Model B, and Model D baselines. The CNN comparison remains a point estimate because only aggregate checkpoint summaries were retained from the original image-model experiments.

enough to matter: macro RMSE 0.765 is promising for coarse ranking, retrieval, or triage, but not convincing enough to replace the VLM teacher when image-level fidelity matters.

4.2 Thresholded Classification View

As an auxiliary analysis, we converted the 1–7 score scale into a binary label: scores ≥ 4 were treated as "affords" and scores ≤ 3 as "does not afford." Table 4 reports positive-class F1 for the reproducible per-image prediction vectors. Under this semantically clean midpoint threshold, Model D remains the strongest overall, with macro F1 0.679. After re-inspecting the boundary examples in Figure 3, we regard this midpoint as the proper semantic cutoff: the score-3 Sleep and Yoga/Stretching cases are visibly kitchens, so treating them as negatives is more defensible than calling them weak positives.

We also swept $\{2.5, 3.0, 3.5, 4.0, 4.5, 5.0, 5.5\}$. Table 5 shows that all reproducible models achieve their best macro F1 at threshold 2.5, with Model D reaching 0.906. We do not report CNN F1 because only aggregate checkpoint summaries, not per-image prediction vectors, were retained from the original image-model experiments.

Figure 3 explains why the lower threshold performs better mechanically. In all three examples, the teacher assigns score 3 and Model D predicts roughly 3.0, so 2.5 preserves them while 4.0 forces them negative. Visual inspection suggests that the stricter decision is the right one for interpretation. The implication is not that 2.5 is the better affordance threshold, but that the students are better calibrated to the teacher's low-middle score range than to the sharper binary boundary at 4. In other words, lowering the threshold improves teacher-matching F1 by relaxing the decision rule, not by producing a better partition.

4.3 Indicator Value, Statistical Power, and Controls

Table 6 reports the paired Model B vs. Model D comparison. Model D achieves lower RMSE on all five affordances, but the statistical interpretation is now stated more carefully. Four improvements are significant under a paired Wilcoxon signed-rank test on absolute errors. The L141

Table 4: Auxiliary thresholded classification analysis. Positive class is defined as VLM score ≥ 4; entries report positive-class F1.

Affordance	Mean	Linear	Model B	Model D
L059 Sleep	0.000	**0.640**	0.578	0.450
L079 Cook	0.000	0.444	0.471	**0.870**
L091 Computer Work	0.000	0.583	0.547	**0.680**
L130 Conversation	0.895	0.882	0.882	**0.932**
L141 Yoga/Stretching	0.000	0.360	0.348	**0.465**
Macro average	0.179	0.582	0.565	**0.679**

Table 5: Best macro F1 from the threshold sweep over $\{2.5, 3.0, 3.5, 4.0, 4.5, 5.0, 5.5\}$. All reproducible models achieved their best macro F1 at threshold 2.5.

Model	Best threshold	Best macro F1
Mean	2.5	0.710
Linear	2.5	0.824
Model B	2.5	0.866
Model D	2.5	0.906

Yoga/Stretching improvement remains numerically meaningful (ΔRMSE $= -0.173$) but is not statistically significant at $\alpha = 0.05$ ($p = 0.0619$). With only 84 test samples per affordance, power to detect small effects is limited; the bootstrap intervals in Figure 4 reflect that uncertainty.

The permutation control adds an important nuance absent from the original draft. On L059, L079, and L091, randomly permuting the indicator assignments largely erases Model D's benefit, which supports the claim that indicator semantics matter. On L130 and L141, however, the permuted model still retains some gain over Model B, so dimensionality and sparsity also contribute. The appropriate claim is therefore not that indicator semantics *fully* explain the gain, but that they explain a substantial fraction of it.

4.4 Feature Ablation, Hyperparameters, and SHAP Matching

The original feature-ablation result remains qualitatively unchanged: presence features explain much of L059 and L079, while aggregate geometry and depth are most helpful for L141. The final LightGBM hyperparameters are reported in Table 7 for reproducibility.

For SHAP analysis, we ranked Model B features by mean absolute SHAP value, tokenized feature names after removing prefixes such as `presence_` and `count_`, and matched them to VLM indicator strings with a word-overlap threshold of 0.3. We interpret the positive SHAP–VLM rank correlations (0.305–0.474 in the original analysis) as suggestive alignment of salient cues, not as evidence that the student has recovered the teacher's causal reasoning.

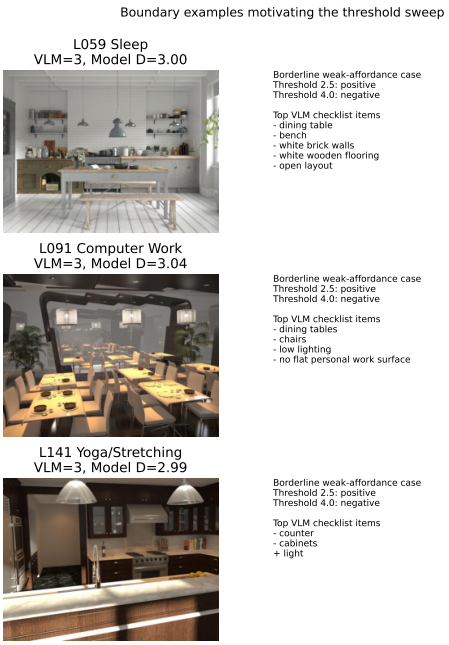


Figure 3: Boundary examples motivating the threshold sweep. Each held-out scene has VLM score 3 and Model D prediction near 3.0, so threshold 2.5 treats it as a weak positive while threshold 4.0 forces it negative. These cases illustrate why the empirically best binary cutoff is lower than the midpoint of the 1–7 scale.

4.5 Score Diagnostics

The VLM score distributions help explain both the successes and the limits of the benchmark. L079 Cook has low entropy (1.19 bits) and a strong floor effect: 74.1% of all scores equal 1. That compression likely makes canonical kitchens easy to separate while also reducing the richness of the target distribution. At the opposite extreme, L059 and L141 have higher entropy and broader support, which is consistent with their greater dependence on layout and mixed cues. Inter-affordance correlations are moderate rather than degenerate (maximum pairwise Pearson $r = 0.462$ between L059 and L141), which suggests the VLM is not simply outputting one generic "good room" score across all tasks.

4.6 Teacher Sanity Checks

Without human labels, the most we can do is ask whether the teacher partitions the scenes in an internally plausible way. Figure 5 provides two sanity checks. First, the threshold-4 co-occurrence matrix shows that some obvious incompatibilities are rare: only 1.7% of images are positive for both Sleep and Cook, and 3.1% for Cook and Yoga/Stretching. Second, conceptually adjacent activities overlap much more, especially Computer Work and Casual Conversation (71.9%), which suggests the teacher often treats lounge-office scenes as multi-functional rather than sharply separable. The high-confidence exemplars are visually plausible for all five affordances. These diagnostics are reassuring enough to justify distillation experi-

Table 6: Paired Model B vs. Model D comparison on the test set. The permutation control preserves indicator dimensionality but shuffles indicator assignments within split, breaking image-indicator semantics.

Affordance	ΔRMSE (D-B)	95% CI	Wilcoxon p	Cohen's d	Permuted D RMSE	Interpretation
L059 Sleep	-0.138	[-0.200, -0.071]	0.0033	0.343	0.815	semantics dominate
L079 Cook	-0.255	[-0.457, -0.067]	< 0.0001	0.484	0.864	semantics dominate
L091 Computer Work	-0.328	[-0.500, -0.154]	0.0038	0.311	1.119	semantics dominate
L130 Conversation	-0.165	[-0.227, -0.103]	< 0.0001	0.539	0.905	mixed: semantics + dimensionality
L141 Yoga/Stretching	-0.173	[-0.288, -0.056]	0.0619	0.229	1.128	mixed: semantics + dimensionality

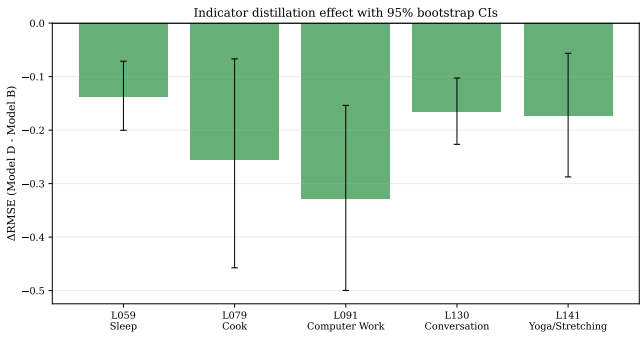


Figure 4: Indicator distillation effect. Bars show ΔRMSE (Model D minus Model B); negative values favor Model D. Error bars are 95% bootstrap confidence intervals over the paired test set.

Table 7: Final LightGBM hyperparameters used to retrain the reported tabular models on train+validation rows.

Model	Aff.	Est.	LR	Leaves	Depth	MinChild
B	L059	276	0.064	91	4	13
B	L079	146	0.068	237	11	6
B	L091	55	0.159	41	4	7
B	L130	486	0.050	219	4	5
B	L141	434	0.184	221	3	26
D	L059	437	0.027	206	4	5
D	L079	194	0.267	142	12	5
D	L091	341	0.052	199	3	5
D	L130	284	0.035	164	3	6
D	L141	401	0.128	176	3	13

ments, but they are still sanity checks rather than evidence that the teacher matches human judgment.

5 Discussion

The evidence supports three claims. First, structured tabular students are effective distillation targets for this teacher on this benchmark. Model B already improves substantially over the best CNN and the mean baseline in four of five affordances, and Model D improves further. Second, the indicator channel provides useful supervision beyond the scalar score, especially on L091 Computer Work where Model B is worse than the mean baseline but Model D recovers to RMSE 0.812 and Pearson $r = 0.595$. Third, the strength of indicator distillation is heterogeneous. On L059, L079, and L091 the random-indicator control collapses back toward Model B, whereas on L130 and L141 some of the gain appears to come from the richer sparse representation itself.

Stronger claims are not defensible here. The paper does not establish true affordance prediction, architecture superiority in general, or real-world transfer. The contribution is a focused study of how well cheap students can approximate a VLM teacher's scene-level scoring behavior under severe data constraints.

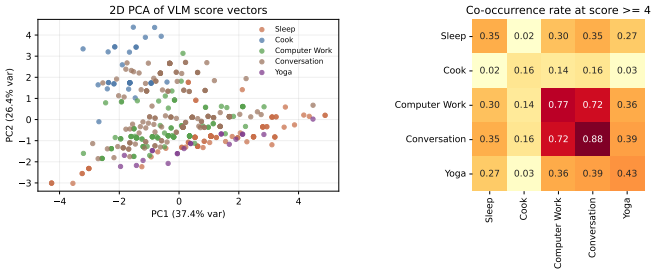
For practical use, the results support deployment only in a narrow sense. Model D is strong enough to act as a cheap front-end for bulk screening, retrieval, or comparison of design alternatives, where occasional score error is acceptable and ambiguous cases can be escalated. The results are not strong enough to justify replacing the VLM teacher outright for image-level analysis. That is the role we envision inside Image_Tagger_3.5, which currently extracts design-relevant visual, spatial, and cognitive attributes from interior images: affordance prediction would be added as another fast analysis layer for cognitive-science and architectural-design workflows, not a final oracle. The threshold sweep reinforces that caution. If one uses the proper semantic threshold of 4, the binary task is harder and the gap between student approximation and categorical judgment becomes more visible. A lower threshold can improve teacher-matching metrics partly by making the classification problem easier.

6 Limitations

Our primary limitation is label validity. The target variable in every experiment is a VLM-derived proxy, not a human affordance judgment. As a result, the reported RMSE, MAE, correlation, and significance tests measure *distillation fidelity* rather than affordance accuracy in the world. The first priority for future work is therefore human validation on at least a subset of scenes. Without that step, the present paper should be read as a study of teacher approximation, not of ground-truth affordance prediction.

The second limitation is domain. All images come from Hypersim, a photorealistic but synthetic dataset. This helps control scene coverage and geometry, but it leaves open the domain gap to real indoor imagery. The segmentation-based feature pipeline may also behave differently on real photographs.

The third limitation is statistical power. Each affordance has only 84 test samples, so confidence intervals remain fairly wide and the study is underpowered for small effects. This



(a) PCA of the five-dimensional VLM score vectors and threshold-4 co-occurrence rates.

(b) High-confidence VLM exemplars for the five affordances.

Figure 5: Internal sanity checks for the VLM teacher. These visualizations test whether the proxy labels behave plausibly; they do not establish human validity.

is visible in L141, where the point estimate favors Model D but the paired test is not significant. The fourth limitation is the uneven reproducibility of retained artifacts: per-image predictions were available for the reconstructed tabular models, but only aggregate checkpoint summaries were retained for the CNNs. We therefore report CNN comparisons transparently as point estimates and avoid unsupported significance claims.

7 Conclusion

This paper is framed around what the experiments actually show. Using 420 synthetic indoor scenes with scene-level splits, lightweight tabular students can approximate Qwen2-VL-7B's affordance scoring behavior better than image-only CNN baselines and simple non-neural baselines. Indicator distillation improves every affordance and is statistically reliable on four of five tasks, while a permutation control shows that semantics explain much, but not all, of the gain. The same ordering also survives a thresholded score-≥ 4 classification view, where Model D attains the strongest macro F1 among the reproducible per-image baselines. These results support compact proxies of large VLM teachers. Establishing whether the same pipeline predicts *human-rated* affordances remains future work.

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